



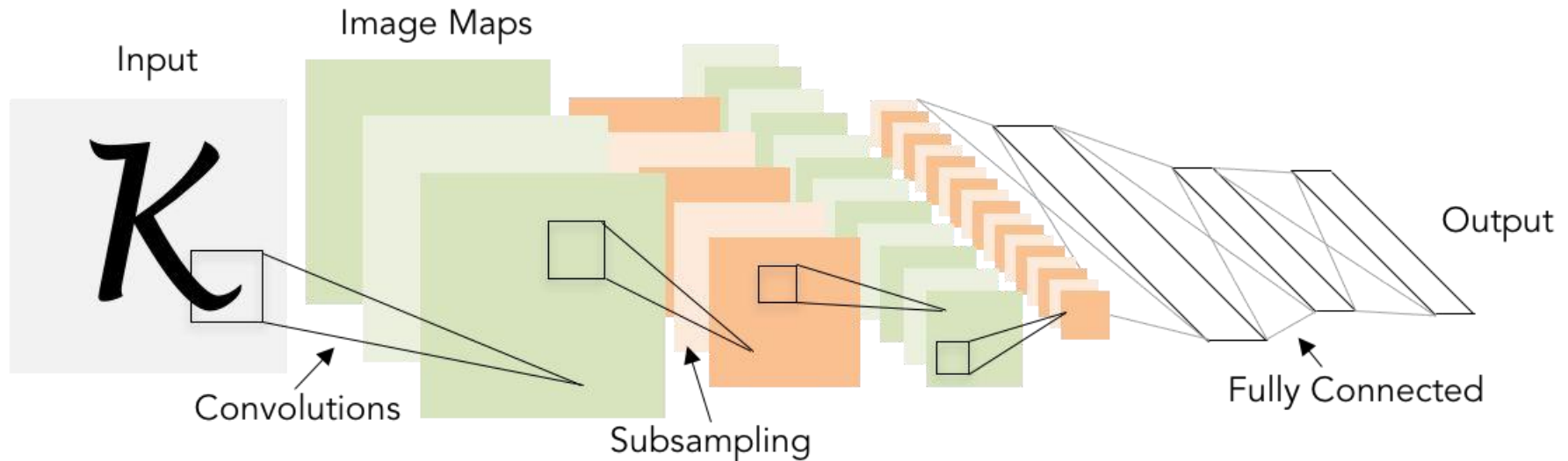
Various CNN Architectures

Content

- **Introduction**
- **VGG**
- **Resnet**
- **1x1 Convolution**

Introduction

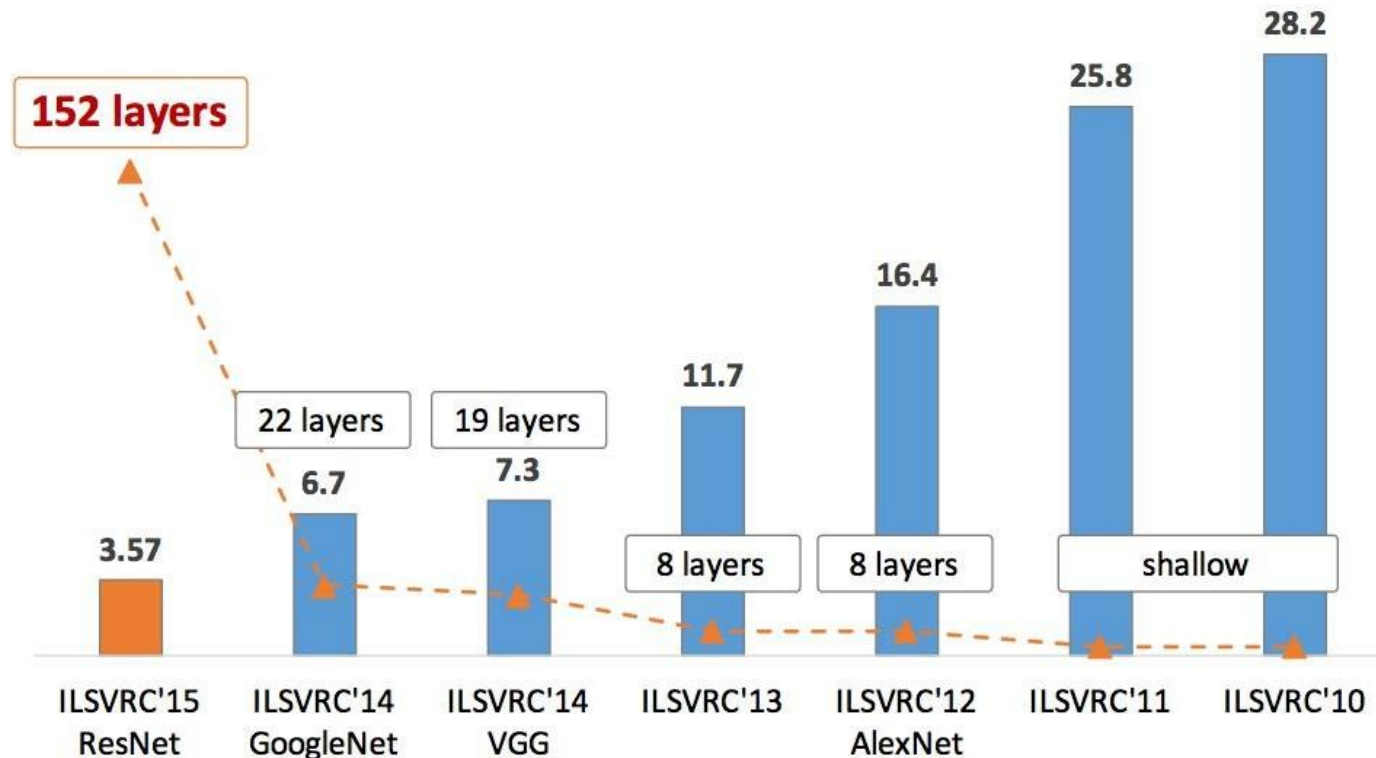
- **LeNet-5**
 - LeCun et al., 1998



Conv filters were 5x5, applied at stride 1
Subsampling (Pooling) layers were 2x2 applied at stride 2
i.e. architecture is [CONV-POOL-CONV-POOL-FC-FC]

Introduction

- ImageNet Large Scale Visual Recognition Challenge (ILSVRC) winners



Introduction

- **ImageNet Dataset**

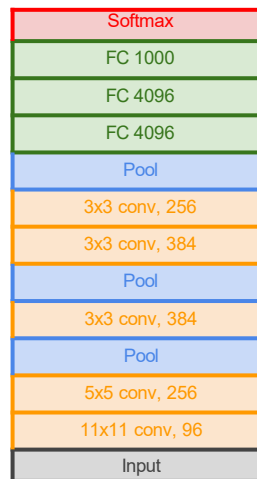
- Images from the real world
- 1000 classes and about 1000 images per class



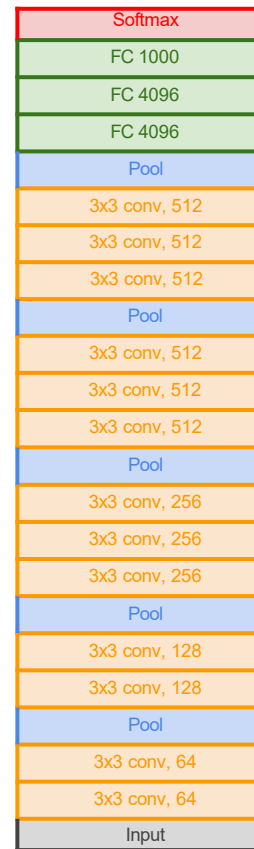
VGGNet

- **Small filters, Deeper networks**

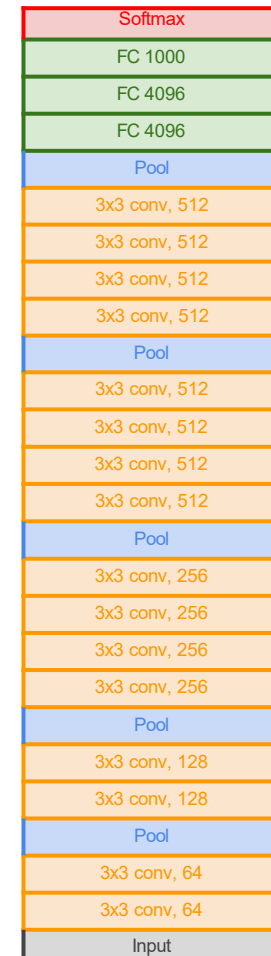
- Only 3x3 CONV stride 1, pad 1 and 2x2 MAX POOL stride 2



AlexNet



VGG16

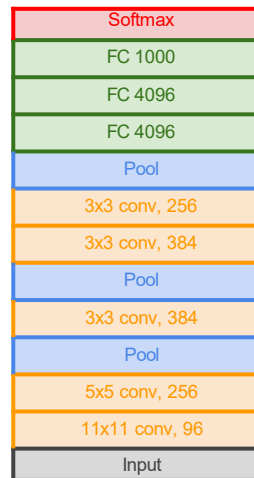


VGG19

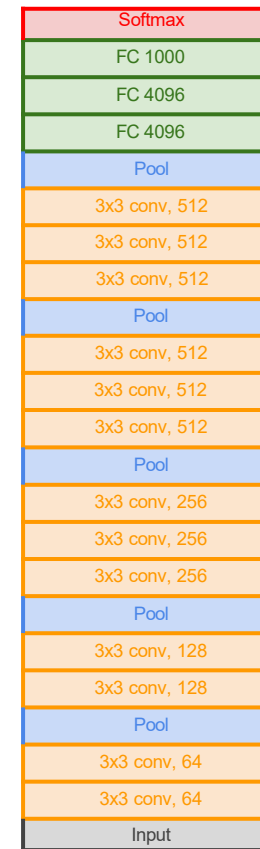
VGGNet

■ Why use smaller filters? (3x3 conv)

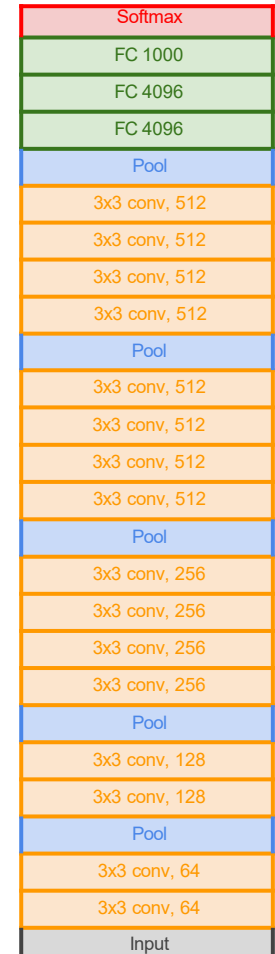
- Three 3x3 conv (stride 1) layers has same **effective receptive field** as one 7x7 conv layer
- Fewer parameters
- Deeper, more non-linearities



AlexNet



VGG16



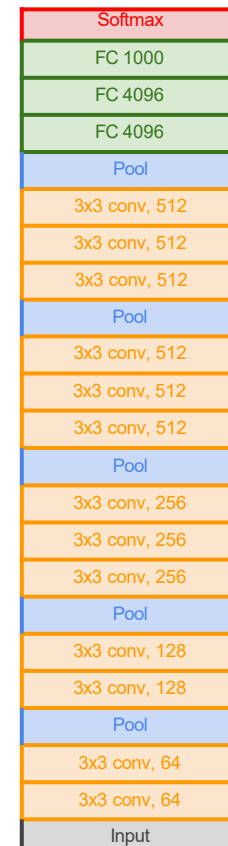
VGG19

VGGNet

INPUT: [224x224x3] memory: $224*224*3=150K$ params: 0
 CONV3-64: [224x224x64] memory: $224*224*64=3.2M$ params: $(3*3*3)*64 = 1,728$
 CONV3-64: [224x224x64] memory: $224*224*64=3.2M$ params: $(3*3*64)*64 = 36,864$
 POOL2: [112x112x64] memory: $112*112*64=800K$ params: 0
 CONV3-128: [112x112x128] memory: $112*112*128=1.6M$ params: $(3*3*64)*128 = 73,728$
 CONV3-128: [112x112x128] memory: $112*112*128=1.6M$ params: $(3*3*128)*128 = 147,456$
 POOL2: [56x56x128] memory: $56*56*128=400K$ params: 0
 CONV3-256: [56x56x256] memory: $56*56*256=800K$ params: $(3*3*128)*256 = 294,912$
 CONV3-256: [56x56x256] memory: $56*56*256=800K$ params: $(3*3*256)*256 = 589,824$
 CONV3-256: [56x56x256] memory: $56*56*256=800K$ params: $(3*3*256)*256 = 589,824$
 POOL2: [28x28x256] memory: $28*28*256=200K$ params: 0
 CONV3-512: [28x28x512] memory: $28*28*512=400K$ params: $(3*3*256)*512 = 1,179,648$
 CONV3-512: [28x28x512] memory: $28*28*512=400K$ params: $(3*3*512)*512 = 2,359,296$
 CONV3-512: [28x28x512] memory: $28*28*512=400K$ params: $(3*3*512)*512 = 2,359,296$
 POOL2: [14x14x512] memory: $14*14*512=100K$ params: 0
 CONV3-512: [14x14x512] memory: $14*14*512=100K$ params: $(3*3*512)*512 = 2,359,296$
 CONV3-512: [14x14x512] memory: $14*14*512=100K$ params: $(3*3*512)*512 = 2,359,296$
 CONV3-512: [14x14x512] memory: $14*14*512=100K$ params: $(3*3*512)*512 = 2,359,296$
 POOL2: [7x7x512] memory: $7*7*512=25K$ params: 0
 FC: [1x1x4096] memory: 4096 params: $7*7*512*4096 = 102,760,448$
 FC: [1x1x4096] memory: 4096 params: $4096*4096 = 16,777,216$
 FC: [1x1x1000] memory: 1000 params: $4096*1000 = 4,096,000$

TOTAL memory: $24M * 4 \text{ bytes} \sim 96MB / \text{image}$ (only forward! ~ 2 for bwd)

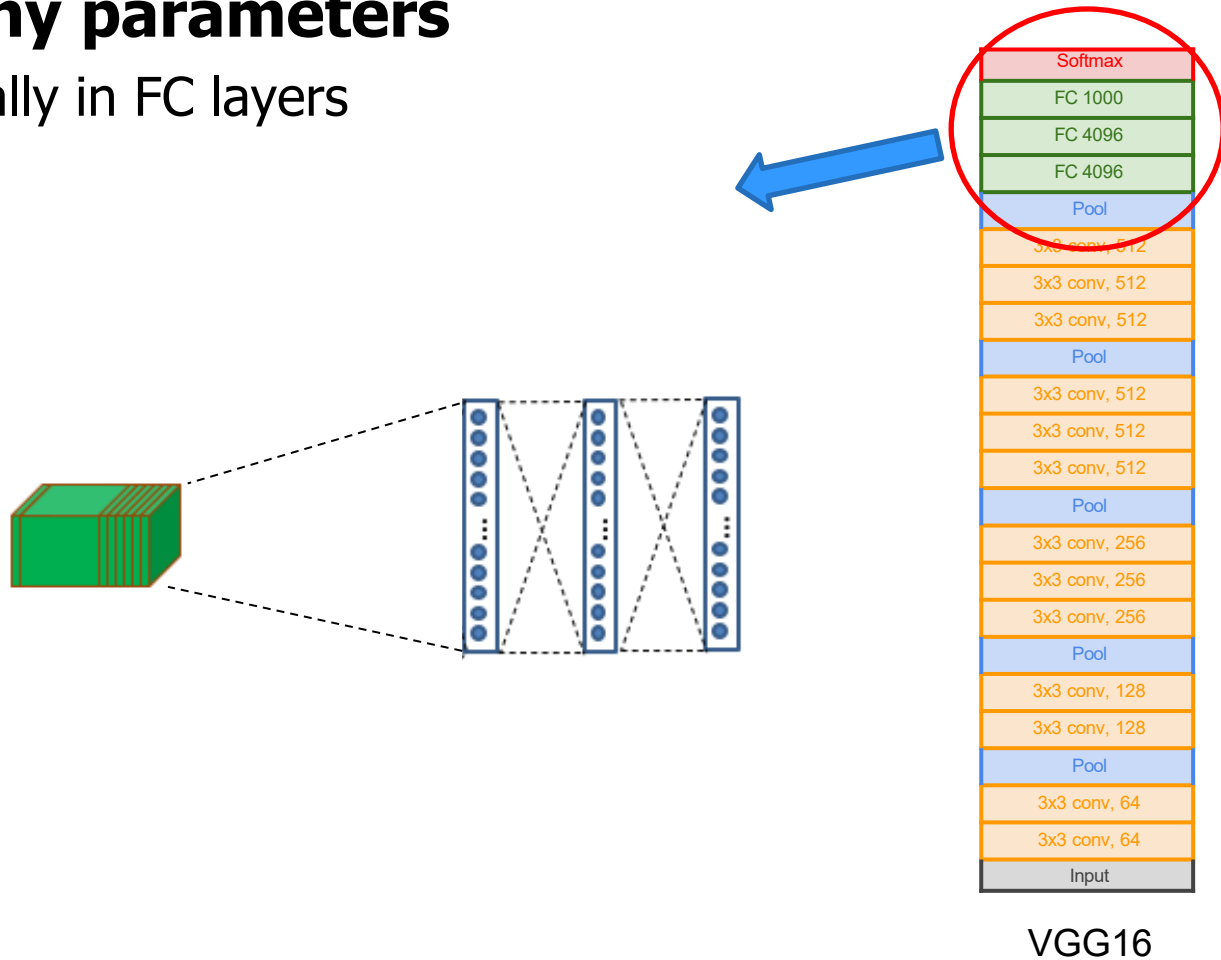
TOTAL params: 138M parameters



VGG16

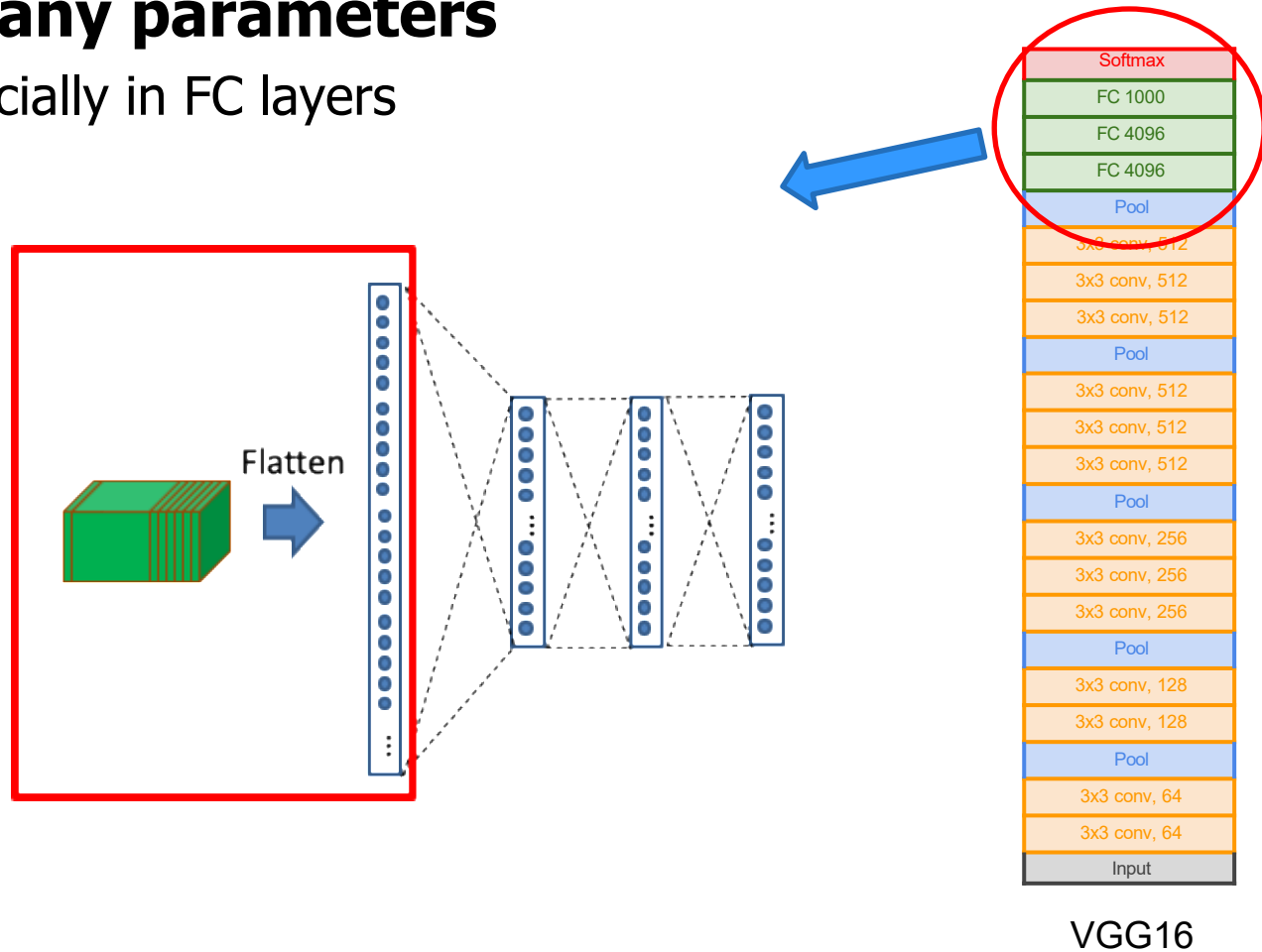
VGGNet

- **Too many parameters**
 - Especially in FC layers



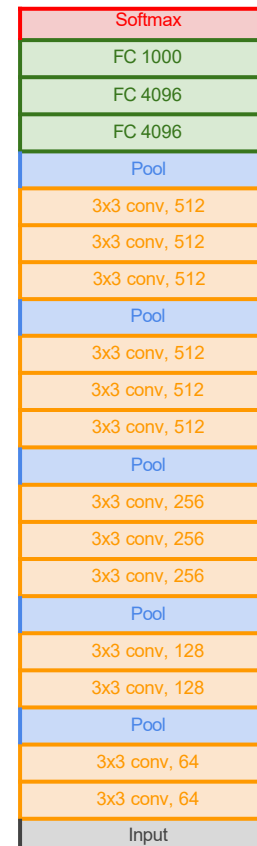
VGGNet

- **Too many parameters**
 - Especially in FC layers

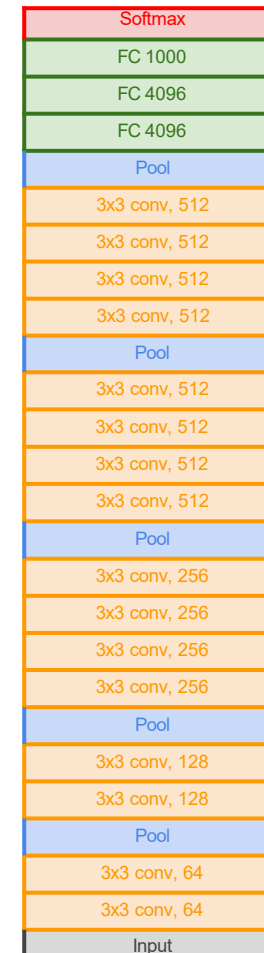


VGGNet

- Only 3x3 filters
- Deeper Structure
- Huge # of parameters



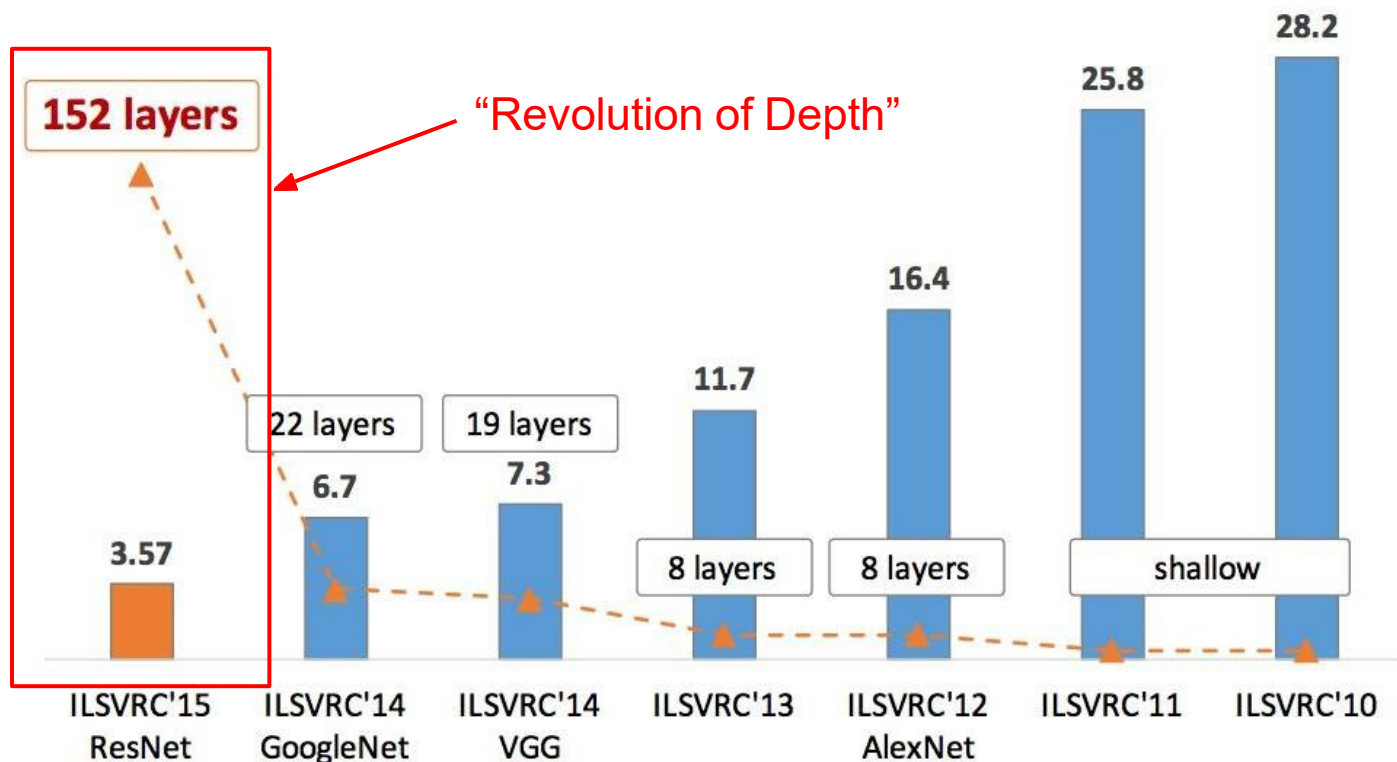
VGG16



VGG19

ResNet

- ImageNet Large Scale Visual Recognition Challenge (ILSVRC) winners



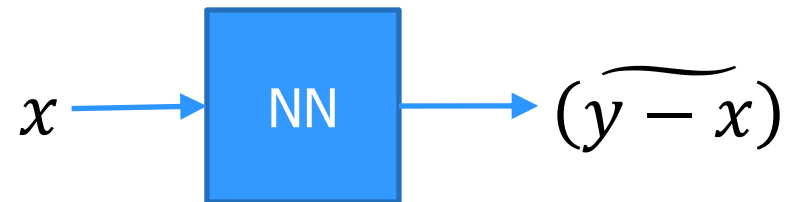
ResNet

X	Y
1.1	1.9
2.0	2.5
3.5	2.9
3.9	4.2



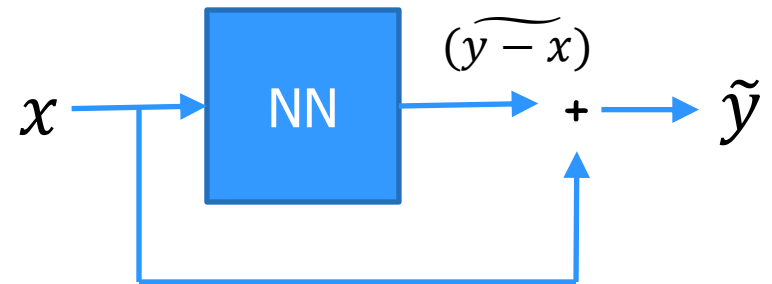
ResNet

X	Y	Y-X
1.1	1.9	0.8
2.0	2.5	0.5
3.5	2.9	-0.6
3.9	4.2	0.3



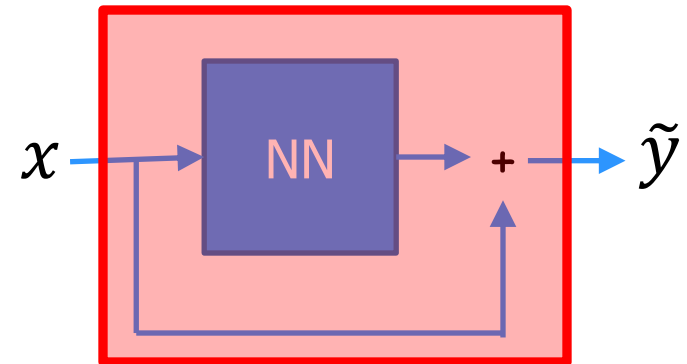
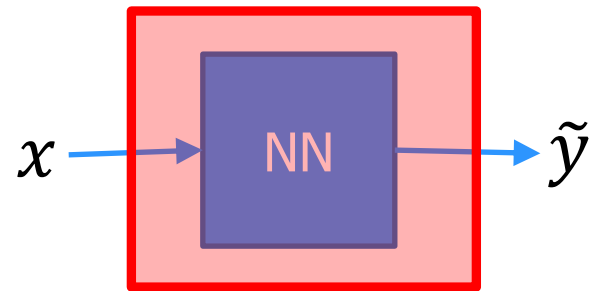
ResNet

X	Y	Y-X
1.1	1.9	0.8
2.0	2.5	0.5
3.5	2.9	-0.6
3.9	4.2	0.3

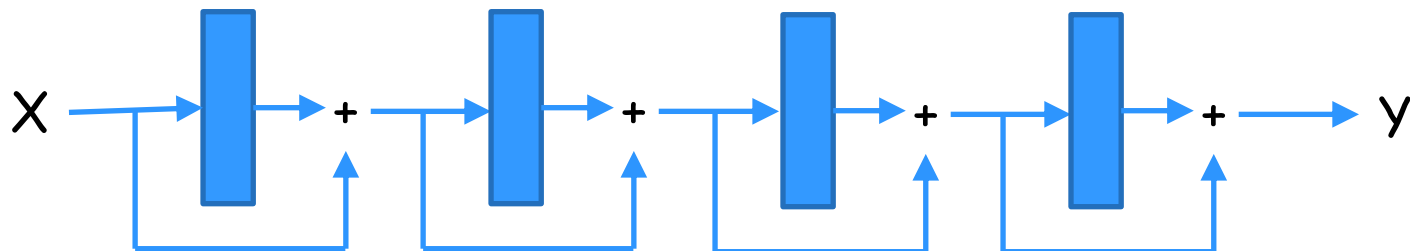
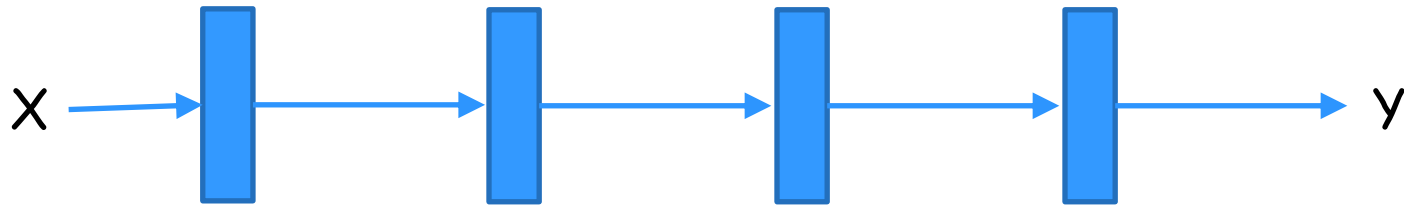


ResNet

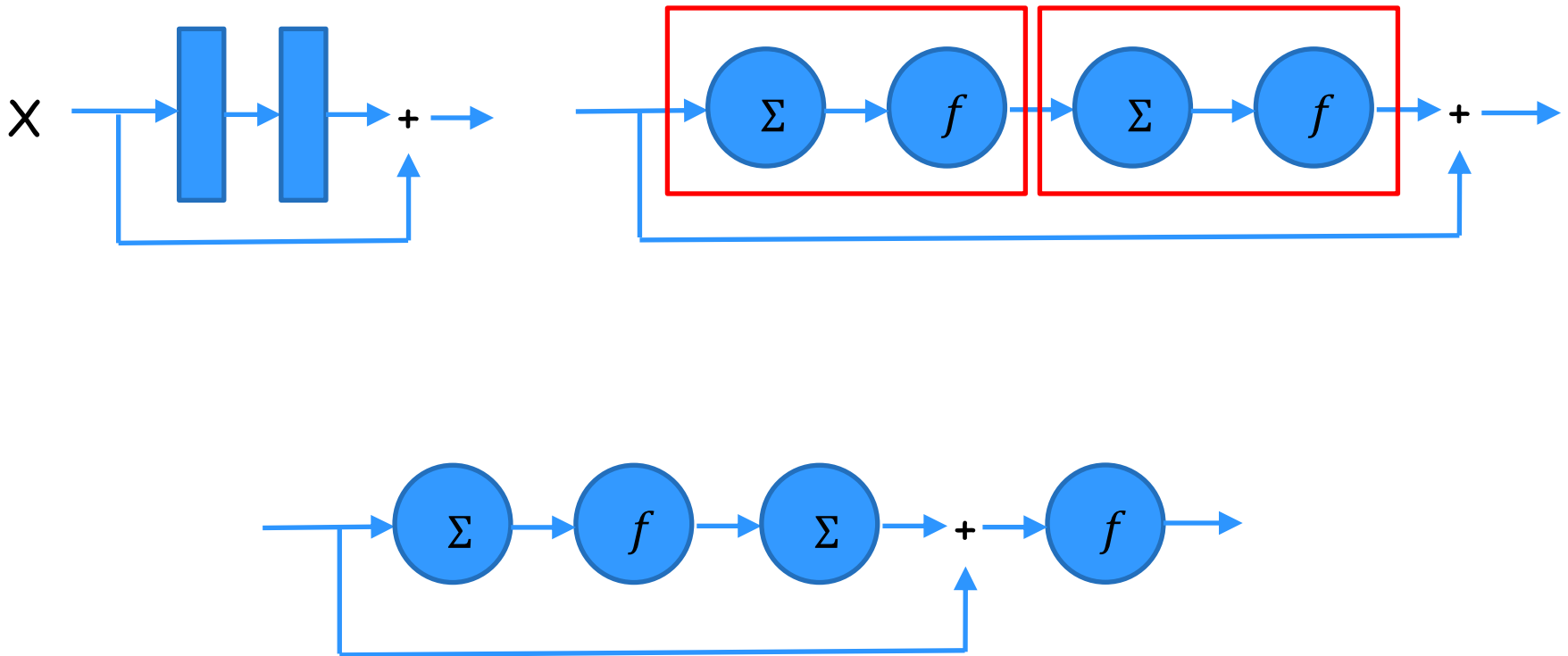
X	Y
1.1	1.9
2.0	2.5
3.5	2.9
3.9	4.2



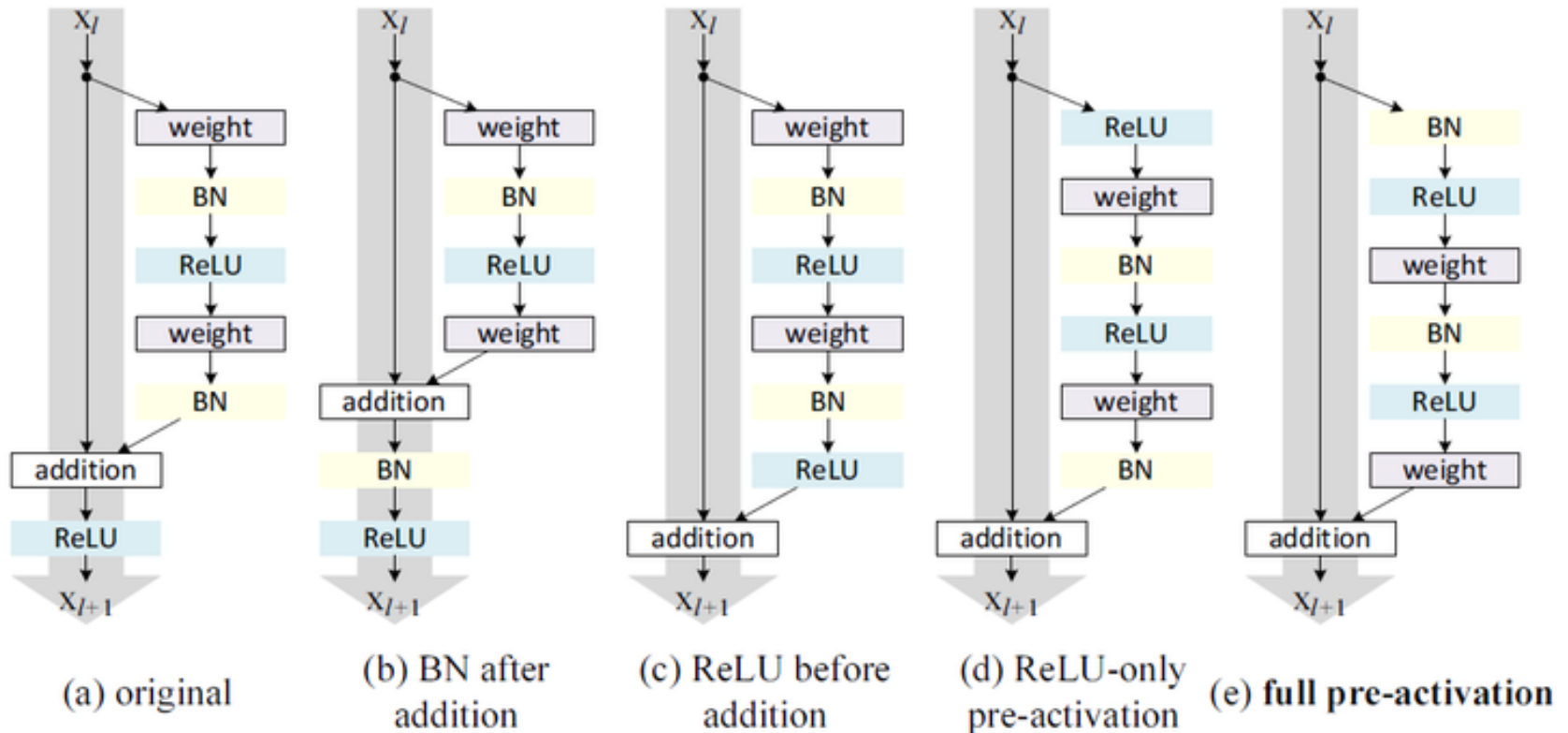
ResNet



ResNet

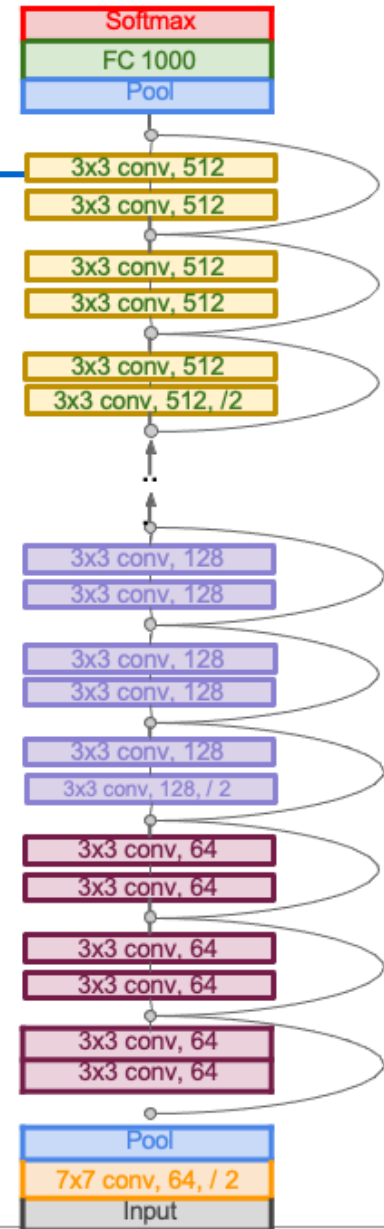
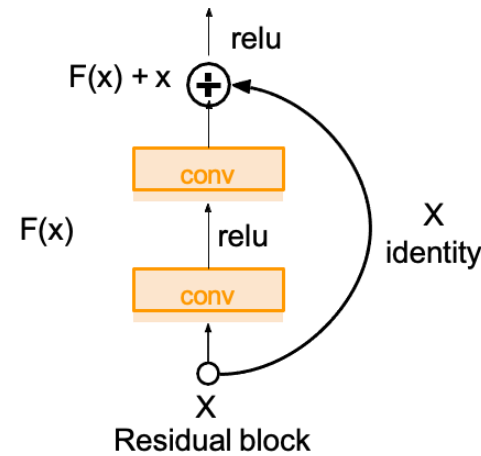
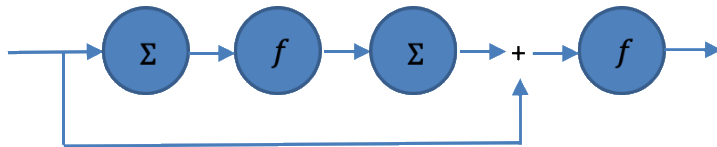


ResNet



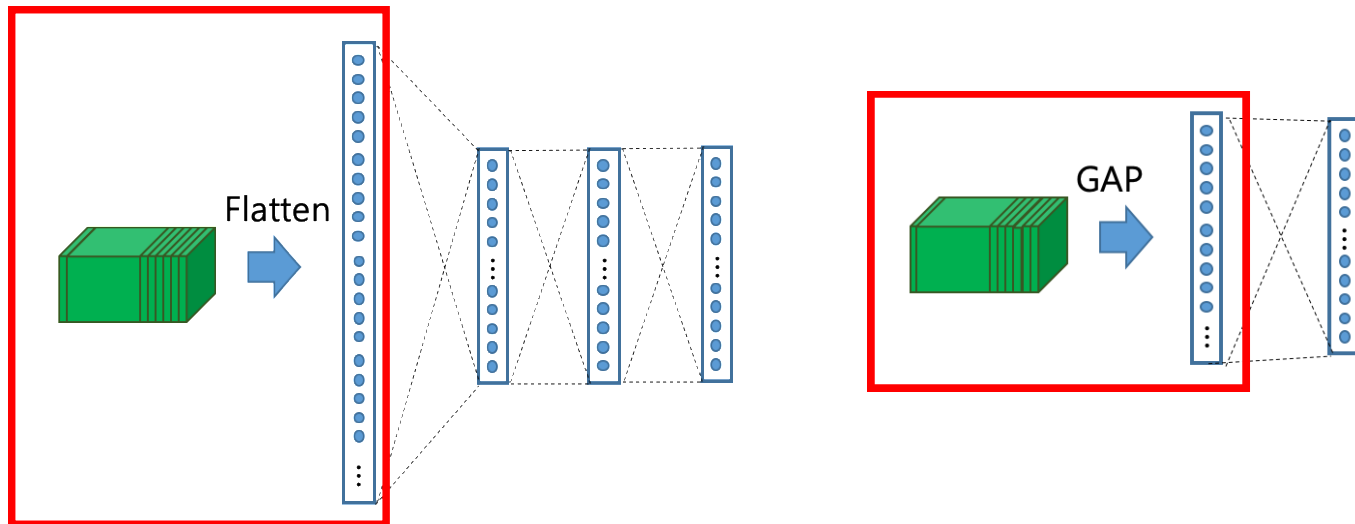
ResNet

- **Very deep networks using residual connections**
 - Every residual block has two 3x3 conv layers
 - Periodically, double # of filters and downsample spatially using stride 2
 - Global average pooling layer after last conv. layer



ResNet

- **GAP vs Flatten**



ResNet

- **Training ResNet in practice:**
 - Batch Normalization after every CONV layer
 - Xavier/2 initialization from He et al.
 - SGD + Momentum (0.9)
 - Learning rate: 0.1, divided by 10 when validation error plateaus
 - Mini-batch size 256
 - Weight decay of $1e-5$
 - No dropout used

Comparing Complexity

